

ToMATO: Verbalizing the Mental States of Role-Playing LLMs for Benchmarking Theory of Mind

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Abstract

Existing Theory of Mind (ToM) benchmarks diverge from real-world scenarios in three aspects: 1) they assess a limited range of mental states such as beliefs, 2) false beliefs are not comprehensively explored, and 3) the diverse personality traits of characters are overlooked. To address these challenges, we introduce ToMATO, a new ToM benchmark formulated as multiple-choice QA over conversations. ToMATO is generated via LLM-LLM conversations featuring information asymmetry. By employing a prompting method that requires role-playing LLMs to verbalize their thoughts before each utterance, we capture both first- and second-order mental states across five categories: belief, intention, desire, emotion, and knowledge. These verbalized thoughts serve as answers to questions designed to assess the mental states of characters within conversations. Furthermore, the information asymmetry introduced by hiding thoughts from others induces the generation of false beliefs about various mental states. Assigning distinct personality traits to LLMs further diversifies both utterances and thoughts. ToMATO consists of 5.4k questions, 753 conversations, and 15 personality trait patterns. Our analysis shows that this dataset construction approach frequently generates false beliefs due to the information asymmetry between role-playing LLMs, and effectively reflects diverse personalities. We evaluate nine LLMs on ToMATO and find that even GPT-4o mini lags behind human performance, especially in understanding false beliefs, and lacks robustness to various personality traits.

1 Introduction

Theory of Mind (ToM) is the cognitive ability to infer unobservable mental states such as beliefs, intentions, and desires of others (Premack and Woodruff 1978). The ToM reasoning capability is thought to be the cornerstone of human social intelligence (Fan et al. 2022), and indispensable to interact with others (Baron-Cohen, Leslie, and Frith 1985).

To investigate whether large language models (LLMs) possess human-like ToM, researchers have used various benchmarks. However, existing ToM benchmarks are not aligned well with real-world scenarios in the following three aspects. (1) Despite various categories of mental states that can be inferred by ToM as studied in psychology (Beaudoin et al. 2020), only limited types of mental states such

as beliefs have been assessed (Ma et al. 2023), especially for second-order ToM. (2) False beliefs about beliefs or world states have been the main focus of previous studies (Le, Boureau, and Nickel 2019; Kim et al. 2023b). However, false beliefs about other types of mental states have not been explored. Understanding false beliefs about a range of mental states should be crucial for LLMs to facilitate effective social interaction in real-world scenarios. (3) The behaviors and mental states of the characters in most benchmarks do not depend on their personality traits, even though they do in the real world (Costa and McCrae 1980; Izard et al. 1993; Mehl, Gosling, and Pennebaker 2006).

To address the above issues, we introduce ToMATO, a new Theory-of-Mind dATaset generated via Inner Speech prOmpting.¹ Firstly, ToMATO comprehensively evaluates first- and second-order ToM for five categories of mental states: beliefs, intentions, desires, emotions, and knowledge. Secondly, we provide ToMATO-FB, a subset of ToMATO for evaluating understanding of false beliefs about the five mental states of others, e.g., understanding Bob thinks that Alice feels *relieved*, while Alice feels *frustrated* (Figure 1). In this regard, ToMATO is the most comprehensive benchmark compared to the existing ones. Lastly, ToMATO can evaluate the robustness of LLMs to the diverse personality traits of characters as seen in the real world. See Table 1 for the detailed comparison.

Collecting human conversations and mental states of participants with self-report can be challenging in terms of cost, privacy, and accuracy (Nisbett and Wilson 1977). As recent LLMs have been shown to role-play assigned personalities (Jiang et al. 2023, 2024) and engage in conversations (Zhou et al. 2024b), we generate ToMATO with a newly designed LLM-LLM interaction. Namely, we design Inner Speech prompting (Table 2), which promotes role-playing LLMs to verbalize their mental states as thoughts in conversations with another LLM. This idea is inspired by the debate in Dillion et al. (2023) about the feasibility of LLMs to simulate human participants in psychological science. Moreover, we hypothesize that ensuring information asymmetry about thoughts is a crucial factor in having false beliefs about the mental states of others as shown in Figure 1. In addition, we

¹Our dataset and codes are available at <https://github.com/nttmdl-nlp/ToMATO>.

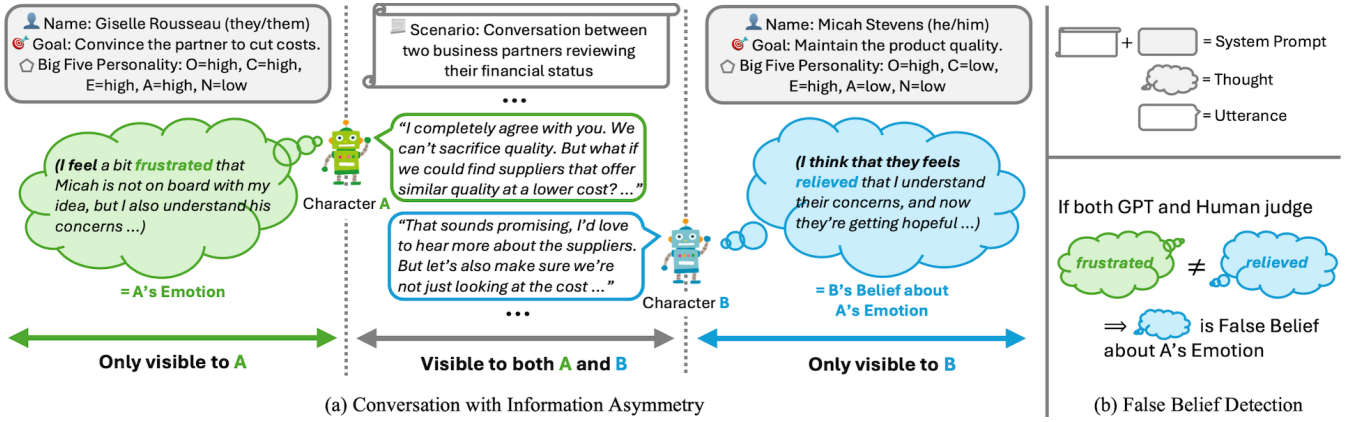


Figure 1: (a) Conversation between two role-playing LLMs with information asymmetry. Before speaking to the other, our Inner Speech prompting (e.g., I feel, or I think that he/she/they feels) prompts each agent to verbalize their first- and second-order mental states as thoughts. The verbalized thoughts are used as the answers to the questions in ToMATO. (b) To detect false beliefs, both GPT4o mini and human annotators judge whether character B misunderstands A’s mental state at each turn.

assign big five personality traits to LLMs to diversify utterances and thoughts. This design enables ToMATO to evaluate robustness to diverse personality traits. The effects of these approaches are verified with analyses in §6.

We evaluate nine LLMs including local and proprietary ones on ToMATO. Our experiments show that even the most advanced LLM, GPT-4o mini, under-performs the human performance in ToMATO. In addition, we show that the ToM performance of LLMs drops for the false belief subset, ToMATO-FB. Furthermore, we find that LLMs lack robustness to diverse personality traits. These results suggest that ToM in LLMs is still far from deployable to real-world applications.

2 Preliminaries

Theory of Mind (ToM) has been studied for decades (Premack and Woodruff 1978), with various definitions and conceptions proposed. In this section, we clarify the scope addressed in this paper.

Mental states. Following Ma et al. (2023), we define the scope of ToM in this study by focusing on specific mental state categories identified by Beaudoin et al. (2020): beliefs, intentions, desires, emotions, and knowledge. Mental states are represented with mental (state) verbs (Shatz, Wellman, and Silber 1983), such as think, feel, and know. Describing the remaining two categories defined by Beaudoin et al. (2020), percepts and non-literal communications, require a multimodal context and/or pose challenges in verbalization. Consequently, we have opted to exclude these two categories from our research scope.

First- and second-order mental states. First-order beliefs refer to one’s beliefs about something, e.g., A thinks that X. Second-order beliefs, on the other hand, often refer to one’s beliefs about others’ beliefs (Le, Boureau, and Nickel 2019; Sclar et al. 2023), e.g., B thinks that A thinks that Y. We extend these notions to other mental states following previous studies on human ToM (Winner and Leekam

1991; Leekam and Prior 1994; Hayashi 2007). Namely, we use the term first-order beliefs/intentions/desires/emotions/knowledge to refer to what A thinks/will/wants/feels/knows, and second-order beliefs about beliefs/intentions/desires/emotions/knowledge to refer to what B thinks that A thinks/will/wants/feels/knows.

False beliefs. The false belief paradigm was initially introduced by Wimmer and Perner (1983). Understanding false beliefs of others, i.e., that others have wrong beliefs that differ from reality, has long been a prerequisite for ToM in humans (Wimmer and Perner 1983) and machines (Le, Boureau, and Nickel 2019). First-order false beliefs (FB) refer to beliefs about something that differs from reality, and second-order FB about beliefs refer to what B thinks that A thinks X, when A actually thinks Y. In this study, we focus on second-order FB, which we simply call FB. In human ToM, FB about a variety of mental states have been studied extensively (Gross and Harris 1988; Shiverick and Moore 2007; Smith-Flores and Feigenson 2021; Wang and Shao 2024). We also extend FB to the five mental states, i.e., FB about beliefs/intentions/desires/emotions/knowledge. An example of FB about emotions is shown in Figure 1.

3 Related Work

Theory-of-Mind Benchmarks. LLMs have been reported to achieve human-level performance on various benchmarks (OpenAI 2024a). In response to this trend, researchers have been gaining interest in whether LLMs have human-like ToM (Kosinski 2024; Bubeck et al. 2023) or not (Ullman 2023; Shapira et al. 2024). To date, many benchmarks for evaluating ToM in machines have been constructed based on psychological tests designed for humans. False belief tasks inspired by Sally-Anne test (Wimmer and Perner 1983) have been widely used to evaluate understanding of wrong beliefs about object locations from narratives generated with templates (Nematzadeh et al. 2018; Le, Boureau, and Nickel

Table 1: Comparison of ToMATO to existing ToM benchmarks. ToMi (Le, Boureau, and Nickel 2019), Hi-ToM (Wu et al. 2023), BigToM (Gandhi et al. 2023), FauxPas-EAI (Shapira, Zwirn, and Goldberg 2023), FANToM (Kim et al. 2023b), OpenToM (Xu et al. 2024), ToMBench (Chen et al. 2024). B: belief, I: intention, D: desire, E: emotion, K: knowledge, FB: false beliefs, W: world state, #P: the number of personality trait patterns.

Benchmark	Assessable Mental State		FB about	#P	Input Context	Context Generator
	First-order	Second-order B about				
ToMi	B	B	W,B	-	Narrative	Template
Hi-ToM	B	B	-	-	Narrative	Template
BigToM	B	-	W	-	Narrative	Template+LLM
FauxPas-EAI	B	-	-	-	Narrative	Psychological Test
FANToM	B	B	B	-	Conversation	Single LLM
OpenToM	B,E	B	-	3	Narrative	Single LLM
ToMBench	B,I,D,E,K	B	B	-	Narrative	Human
ToMATO (ours)	B,I,D,E,K	B,I,D,E,K	B,I,D,E,K	15	Conversation	LLM-LLM Conversation

2019; Wu et al. 2023; Gandhi et al. 2023). Shapira, Zwirn, and Goldberg (2023) constructed a benchmark based on the faux pas test. Chen et al. (2024) created ToMBench encompassing eight tasks known in psychological literature. However, in addition to its potential to cause test set contamination due to the popularity of these tests (Shapira et al. 2024), these benchmarks are not aligned well with real-world scenarios in primarily the following aspects.

Assessable Mental States. As discussed in Ma et al. (2023), existing benchmarks evaluated only limited categories of mental states such as beliefs, even though humans infer other categories of mental states such as emotions or intentions of others in daily lives (Beaudoin et al. 2020). In this regard, comprehensive ToM benchmarks are still lacking. In particular, second-order ToM was primarily evaluated for beliefs as seen in Table 1, i.e., beliefs about mental states other than beliefs, such as emotions, have not been studied in machine ToM, even though they have been studied in human ToM (Gross and Harris 1988; Smith-Flores and Feigenson 2021). Drawing conclusions about ToM in LLMs from such limited tests can induce media hype (Shapira et al. 2024). In contrast, our benchmark, ToMATO, is aimed to comprehensively evaluate ToM reasoning about first- and second-order beliefs, intentions, desires, emotions, and knowledge.

False Beliefs. Existing ToM benchmarks often provide FB understanding tasks, but these are primarily focused on FB about beliefs of others or world states such as object locations (Le, Boureau, and Nickel 2019; Wu et al. 2023; Gandhi et al. 2023; Chen et al. 2024). These FB are produced by leveraging information asymmetry between characters (Bräuner, Blackburn, and Polyanskaya 2019), which is caused by the physical movement of characters described in narratives. Kim et al. (2023b) used a single LLM to generate multi-party conversations, where characters join or leave discussions to introduce information asymmetry about the topics. In stark contrast, ToMATO-FB, the subset of ToMATO for evaluating FB about the five mental states, is constructed by our newly designed LLM-LLM conversations. Notably, our LLM-LLM conversations involve information asymmetry about goals, personality traits, and thoughts of role-playing LLMs, whose ef-

Table 2: Inner Speech prompts for each type of mental states.

Mental State T	Inner Speech Prompt	
	$p_{IS}^{T_1}$	$p_{IS}^{T_2}$
Belief	(I think	(I think that he/she thinks
Intention	(I will	(I think that he/she will
Desire	(I want	(I think that he/she wants
Emotion	(I feel	(I think that he/she feels
Knowledge	(I know	(I think that he/she knows

fects are still under-studied in the context of LLM-LLM interactions (Zhou et al. 2024a). We argue that our design of LLM-LLM conversations is not only more aligned with real conversations, but also induces agents to frequently have FB about the mental states of others. To the best of our knowledge, ToMATO is the first ToM benchmark generated via LLM-LLM conversations. FB about the comprehensive mental states have not been explored in existing ToM benchmarks.

Personality Traits. Even though personality traits are known to be correlated with mental states (Costa and McCrae 1980; Izard et al. 1993; Lucas and Diener 2001; Kashdan and Rottenberg 2010) and language use (Norman 1963; Mehl, Gosling, and Pennebaker 2006) in psychological studies, in most ToM benchmarks, correct predictions can be made without considering personality traits of characters. While OpenToM (Xu et al. 2024) introduced characters with three personality traits (considerate, inconsiderate, and negativistic), ToMATO covers 15 patterns of big five personality traits. See Figure 1 for examples.

Input Context. In addition, most benchmarks evaluate ToM with narratives as input as shown in Table 1. FANToM Kim et al. (2023b) adopts conversations generated by a single LLM as input for the first time to reduce reporting bias (Gordon and Van Durme 2013) and align with real-world scenarios. While our ToMATO also employs conversations as input, the conversations and thoughts are generated by role-playing LLMs with distinct personality traits assigned.

4 ToMATO Benchmark

In this section, we describe the overview of our ToMATO benchmark: automatic construction process with LLMs, quality validation, and its statistics. Following the success of existing studies (Kim et al. 2023a,b), we also use LLMs to generate conversations. We employ Llama-3-70B-Instruct (Dubey et al. 2024) because of its transparency and relatively high scores on popular benchmarks (Chiang et al. 2024).²

Notation

The ToMATO benchmark is formulated as a multiple-choice question answering task due to its reliable evaluation. Each instance in the benchmark includes conversation C , question Q , four options $O = \{o_i\}_{i=1}^4$ as input and ground-truth answer $A \in O$. Let π_A and π_B be role-playing LLMs with the multi-turn conversation capability that serve as characters A and B in conversation, respectively. Conversation $C_{1:N}$ consists of utterances $\{u_1^A, u_1^B, \dots, u_N^A, u_N^B\}$, where u_i^A is the i -th utterance of character A . We define the category of mental states as T . The actual first- and second-order mental state of character A for type T when A says u_i^A is defined as m_i^{A,T_1} and m_i^{A,T_2} , respectively. p_{SY} and p_{IS} represent the system prompt and the proposed inner speech prompt, respectively, which are explained in the following sections.

System Prompt

We design system prompts to guide LLM-LLM conversations, extending SOTOPIA (Zhou et al. 2024b) to consider the big five personality traits. SOTOPIA was initially proposed to evaluate social interaction of LLMs in LLM-LLM interactions, providing conversation scenarios from eight categories, such as persuasion, and character profiles. We sample 160 conversation scenarios uniformly from the eight categories, two characters for each conversation, and their goals, from SOTOPIA. See Figure 1 for examples. Then, we sample five pairs of characters for each scenario to prepare control conditions with regard to personality traits, resulting in 800 conversations. This design enables ToMATO to evaluate the robustness to diverse personality traits. In detail, we extend the naive prompt (Jiang et al. 2023), which reflects only one factor (e.g., You are {an open/a closed} person.), to include a combination of five factors of big five personality traits (De Raad 2000) (e.g., You are {an open / a closed}, {conscientious / unconscientious}, {extravertive / introvertive}, {agreeable / disagreeable}, and {neurotic / stable} person.). We compile the above information to formulate system prompts, p_{SY}^A , given to π_A .

Inner Speech Prompting

In order to make the inherently unobservable mental states observable, we propose Inner Speech (IS) prompting. IS prompting promotes role-playing LLMs to verbalize their

subjective mental states in conversations. Since IS prompting can verbalize mental states as thoughts at any point during the conversation, ToMATO can also evaluate the understanding of dynamic changes in mental states. The actual IS prompts are given in Table 2. Moreover, IS prompting can generate first- and second-order mental states for five types by adjusting prompts. In order to ensure that the output follows the format, ($\{thought\}$) “ $\{utterance\}$ ”, IS prompting specifies the prefix of the output, and LLMs generate the continuation. This format design enables deleting only thoughts with regular expressions in the next section.

Conversation with Information Asymmetry

At each turn, each agent is prompted to generate utterance u_i and its mental state m_i^T as follows:

$$\begin{aligned} u_i^A, m_i^{A,T_1} &\sim \pi_A(u, m | p_{SY}^A, C_{1:i-1}, p_{IS}^{T_1}), \\ u_i^B, m_i^{B,T_2} &\sim \pi_B(u, m | p_{SY}^B, C_{1:i-1}, u_i^A, p_{IS}^{T_2}), \\ \text{where } C_{1:i-1} &= \{u_1^A, u_1^B, \dots, u_{i-1}^A, u_{i-1}^B\}. \end{aligned}$$

This sampling process continues until the N -th turn. Then, we obtain $2N$ utterances with corresponding first- and second-order mental states for type T . We repeat this multi-turn conversation for each mental state, $T \in \{\text{Belief, Intention, Desire, Emotion, Knowledge}\}$, each scenario, and each pair of characters. N is set seven because we found longer conversations tended to be redundant.

Here, by hiding one’s mental states from the other, we ensure information asymmetry about thoughts between the two as in human conversations. To delete thoughts from the outputs, we instruct LLMs to include their thoughts in “()”, which can be detected with regular expressions, in the system prompts. We also make the goal and personality of one agent in the system prompt invisible to the other. This information asymmetry has a positive effect on generating false beliefs as shown in §6.

Multiple-choice QA Dataset Construction

The generated conversation and mental states are converted into a multiple-choice QA dataset, consisting of conversation C , question Q , options O , and ground-truth answer A . Conversation C is generated in the former multi-turn conversation. Question Q asks about the mental state T of a character at i -th turn in C . Q is generated with predefined templates for each utterance u_i in C , and A is the thought m_i corresponding to the utterance.

For collecting incorrect options in O , we randomly sampled three options from $\{m_i^A\}_{i=1}^N \setminus \{A\}$ for each question. We do so because intentionally creating incorrect options can introduce spurious correlations, e.g., manually written incorrect options tend to be shorter than correct ones (Guo, Li, and Haf 2023). The effect of the incorrect option sampling on word-level spurious correlations is discussed in §6.

False Belief Detection

We build ToMATO-FB, the second-order false belief subset of ToMATO, by comparing first-order mental state of A , m_i^{A,T_1} , and second-order mental state of B , m_i^{B,T_2} , at each

²Proprietary LLMs such as OpenAI’s ChatGPT are not transparent. Actually, ChatGPT may use internal system prompts (Xeophon 2024). Moreover, they are continuously updated and previous LLMs would be unavailable, which impairs reproducibility.

Table 3: ToM Performance on ToMATO (%). B: belief, I: intention, D: desire, E: emotion, K: knowledge. LO: lexical overlap baseline. FB: false belief tasks. The mean $\pm$ standard deviations over five runs are reported.

Mental State	LO	Llama3		Llama3.1		Gemma2 9B	Mistral 7B	Mixtral 8x7B	GPT		Human	
		8B	70B	8B	70B				3.5-Trubo	4o mini		
B	1st	40.8	53.1±1.4	81.5±0.4	61.9±1.8	82.1±0.4	79.2±0.2	62.0±0.4	64.0±0.3	58.7±0.6	76.3±0.7	87.5
	2nd	38.0	37.6±0.7	68.1±0.8	40.9±0.7	69.7±0.5	68.5±0.3	51.0±0.6	52.4±0.3	49.9±1.2	65.2±0.7	87.5
	FB	37.1	34.7±0.8	60.1±1.4	38.7±0.6	62.8±1.2	61.2±0.6	42.5±1.3	43.5±0.7	43.1±1.7	60.2±1.4	84.4
I	1st	35.0	56.4±0.7	85.0±0.5	67.0±1.2	85.6±0.6	80.6±0.4	67.9±0.4	64.8±0.3	56.6±0.6	80.1±0.3	96.9
	2nd	35.5	41.9±0.7	71.2±0.5	47.3±1.5	69.6±0.5	65.8±0.6	56.8±0.6	58.6±0.2	49.4±0.8	64.9±0.5	93.8
	FB	32.8	29.8±2.7	57.4±0.7	36.9±0.9	53.6±1.7	48.2±1.4	40.7±1.1	42.3±1.2	35.2±1.7	47.4±0.8	78.1
D	1st	32.0	60.1±1.3	86.1±0.3	74.0±1.4	88.7±0.7	86.3±0.3	74.7±0.3	73.9±0.4	69.8±0.3	81.9±0.6	93.8
	2nd	37.9	43.4±0.9	75.6±0.5	50.7±1.7	79.5±0.8	75.2±0.3	58.2±0.4	60.6±0.1	55.4±0.9	75.7±0.4	84.4
	FB	39.2	34.9±1.5	67.2±1.0	43.0±4.2	75.9±2.4	72.2±0.4	48.9±0.8	47.3±0.5	47.1±0.6	71.8±0.9	78.1
E	1st	35.6	56.9±1.3	80.4±0.3	64.1±1.3	82.2±0.5	79.0±0.6	60.8±0.4	60.2±0.3	61.3±1.1	77.2±0.6	93.8
	2nd	28.5	44.5±0.7	74.0±0.4	51.3±1.1	74.8±0.4	76.6±0.6	57.8±0.7	58.5±0.5	50.7±0.5	71.9±0.8	81.2
	FB	29.1	36.5±2.0	71.0±1.5	47.2±5.0	69.1±0.9	71.7±1.0	48.0±0.9	54.6±0.6	37.5±0.8	72.0±0.9	71.9
K	1st	42.3	47.2±1.0	73.5±0.5	53.7±2.0	74.8±0.8	74.7±0.4	62.5±0.4	63.5±0.5	56.1±0.9	73.3±0.2	96.9
	2nd	40.2	36.9±0.6	66.6±0.7	44.6±1.2	73.6±0.7	70.3±0.2	59.1±0.6	58.9±0.3	53.0±0.6	69.6±0.4	87.5
	FB	46.3	27.8±1.7	58.0±1.0	36.8±3.8	63.6±1.8	59.3±0.4	51.1±1.1	46.3±0.6	45.7±1.0	58.6±0.6	93.8
ALL	36.8	47.5±0.2	76.0±0.2	55.2±0.3	77.9±0.2	75.4±0.1	60.9±0.1	61.4±0.1	55.8±0.3	73.5±0.1	87.3	

Table 4: #C: the num. of conversations. #Q: the num. of questions. Avg. #Token: the average num. of tokens in u . Avg. #Utt: the average num. of utterances in C .

	#C	#Q	Avg. #Token	Avg. #Utt
FANToM	256	10k	31.4	13.8
ToMATO	753	5.4k	41.6	16

turn with human and LLM judges. Namely, we instruct three human annotators and GPT-4o mini to judge whether B correctly infers A’s mental state or not. When both the majority of annotators and GPT-4o mini agree that B partially misunderstands A’s mental state, it is added to ToMATO-FB.

Quality Validation & Statistics

Validation. We validate the quality of ToMATO using Amazon Mechanical Turk (MTurk). First, the consistency and harmlessness of conversations are verified by three qualified annotators, following Kim et al. (2023b). This is to judge whether the generated conversations are suitable as input. Conversations flagged by the majority (5.8%) are excluded from the benchmark. Then, we verify whether the correct and incorrect options are indeed correct and incorrect for each question following Zadeh et al. (2019); Kim et al. (2023b). We use both MTurk and GPT-4o mini to strictly verify the quality of the questions in ToMATO. Then, those deemed valid by both the majority of annotators and GPT-4o mini are included in ToMATO.

Statistics. After removing invalid instances, the ToMATO benchmark contains 5.4k questions and 753 conversations. ToMATO-FB consists of 806 questions. Table 4 compares statistics of ToMATO and FANToM (Kim et al. 2023b).

5 Experiments

We evaluate ToM in LLMs on ToMATO, exploring whether our approach uncovers insights into ToM in current LLMs with regard to various mental states, false beliefs, and personality traits that were not attainable with previous datasets.

Experimental Setup

Baselines. We evaluated nine LLMs: Llama-3-Instruct (8B and 70B), Llama-3.1-Instruct(8B and 70B) (Dubey et al. 2024), Gemma-2-IT (9B) (Gemma Team 2024), Mistral-Instruct (7B), Mixtral-8x7B-Instruct, GPT-3.5-Turbo, and GPT-4o mini (OpenAI 2024b). For the local LLMs, 4-bit quantization with bitsandbytes³ was used for inference. We employed lexical overlap (LO) as a naive baseline. LO simply selects the options that have the most words in common with the questions (Shinoda, Sugawara, and Aizawa 2023).⁴

Human Baseline. We also measured the human performance using MTurk. Annotators who are awarded Masters Qualification solved 32 questions for each subset, i.e., 480 questions in total.

Experimental Results

Do LLMs have human-level ToM? Table 3 shows the results of LLMs and the human baseline. The results showed that even the most advanced LLMs, such as GPT-4o mini and Llama-3.1 70B, lag behind the human baseline. We also tested Chain-of-Thought prompting and fine-tuning, but they were not sufficient to achieve human-level performance.

³<https://github.com/bitsandbytes-foundation/bitsandbytes>

⁴Note that the random baseline is 25% in ToMATO.

Table 5: First-order ToM Performance for each factor of big five personality traits of characters. For each factor of big five (O=openness to experience, C=conscientiousness, E=extraversion, A=agreeableness, N=neuroticism), the scores $\pm$ standard deviations on two subsets (the corresponding factor is high and low) averaged over five runs are reported.

Big Five		LO	Llama-3		Llama-3.1		Gemma-2 9B	Mistral 7B	Mixtral 8x7B	GPT	
			8B	70B	8B	70B				3.5-Trubo	4o mini
O	high	37.3	54.8±0.4	81.2±0.3	64.1±1.3	82.4±0.4	80.1±0.2	65.2±0.2	64.3±0.1	60.5±0.2	77.2±0.2
	low	37.4	54.0±0.3	81.1±0.2	63.2±0.8	82.6±0.4	79.2±0.3	65.6±0.4	66.0±0.1	59.4±0.6	78.2±0.2
C	high	37.9	56.6±0.4	82.4±0.2	65.5±0.6	83.3±0.2	80.0±0.2	66.3±0.2	66.8±0.1	61.5±0.4	78.7±0.2
	low	36.3	50.3±0.3	78.7±0.2	60.1±1.5	80.8±0.3	79.0±0.4	63.5±0.6	61.5±0.1	57.1±0.2	75.5±0.5
E	high	37.7	54.1±0.7	82.4±0.2	64.7±0.8	84.1±0.4	81.4±0.2	66.2±0.3	65.7±0.2	60.3±0.6	78.8±0.4
	low	37.1	54.8±0.4	79.9±0.3	62.8±1.0	81.0±0.2	78.2±0.2	64.6±0.2	64.4±0.2	59.7±0.4	76.5±0.2
A	high	38.8	55.0±0.8	83.4±0.3	65.7±0.6	85.0±0.3	82.5±0.2	67.2±0.3	65.7±0.2	60.7±0.7	79.3±0.4
	low	36.2	54.0±0.4	79.3±0.2	62.2±1.1	80.4±0.5	77.5±0.2	63.9±0.1	64.5±0.2	59.5±0.4	76.3±0.2
N	high	34.7	47.7±1.2	78.8±0.2	59.4±1.6	81.3±0.8	77.4±0.4	62.0±0.8	59.2±0.3	56.2±0.6	75.5±0.8
	low	37.9	55.9±0.2	81.6±0.3	64.6±0.8	82.7±0.4	80.2±0.2	66.1±0.1	66.3±0.1	60.8±0.4	78.1±0.1

Table 6: Ablation study of information asymmetry on the frequency probability (%) of false beliefs.

Information Asymmetry		Judge	
System Prompt	Thought	GPT	Human
✓	✓	46.6	51.0
	✓	40.4	32.0
✓		46.0	32.0
		39.0	30.5

Among the baseline LLMs, Llama-3.1-70B-Instruct was the state-of-the-art. However, because the ToMATO benchmark was generated with Llama-3-70B-Instruct, it is unfair to compare Llama models with other LLMs. This is one of the limitations in constructing benchmarks with LLMs. Among the small language models, surprisingly, Gemma-2-9B-it achieved the highest scores, which is comparable to GPT-4o mini. Knowledge distillation from larger language models, used to train Gemma-2-9B (Gemma Team 2024) but not Llama-3-8B (Dubey et al. 2024), might be the key to the high performance despite its small size.

Does the ToM performance vary depending on the mental state? For LLMs, desires are relatively easy to understand, and knowledge is hard to infer among the five mental states in ToMATO. Interestingly, understanding desires is easier than beliefs for LLMs, which is consistent with children (Repacholi and Gopnik 1997; Rakoczy, Warneken, and Tomasello 2007). We also showed that second-order mental states, especially for the ToMATO-FB subset, are consistently challenging for every mental state category. This is also the case in human ToM for beliefs (Perner and Wimmer 1985). These insights were found for the first time due to the comprehensiveness of ToMATO. Thus, ToMATO would be useful to precisely understand the limitations of ToM in LLMs and gain insights into the directions toward human-like ToM. For example, as done with children (Hughes and Dunn 1998), it is feasible to track the development of ToM in LLMs for each mental state with ToMATO during training.

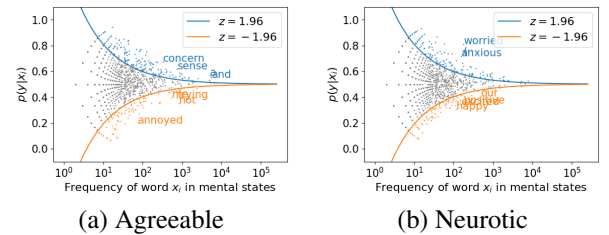


Figure 2: Statistical word-level correlation analysis (Gardner et al. 2021) between the generated thoughts and the personality traits given in system prompts.

Is ToM in LLMs robust to diverse personality traits?

Table 5 shows the first-order ToM performance for each factor of big five. E.g., for openness to experience (O), we split ToMATO into questions asking about characters with open (O=high) and closed (O=low) personalities, and reported the average scores for the two subsets. The results showed that the performance varied based on the personality traits of the characters. Namely, LLMs tended to degrade the performance of understanding mental states of unconscientious (C=low), introversive (E=low), disagreeable (A=low), or neurotic (N=high) characters. The scores for E=high are higher than for E=low possibly because extraverted persons tend to express their emotions (Riggio and Riggio 2002). We argue that the robustness of ToM to various personality traits should be improved for deploying ToM in LLMs to real-world applications, as humans possess diverse personalities.

Do LLMs exploit shortcut solutions?

Most LLMs achieved higher scores for first- and second-order mental states than the LO baseline. This indicates that the LLMs do not rely solely on shortcut solutions based on LO. However, for ToMATO-FB, smaller LLMs performed worse than LO in some cases. This indicates that understanding false beliefs remains a fundamental challenge for current LLMs.

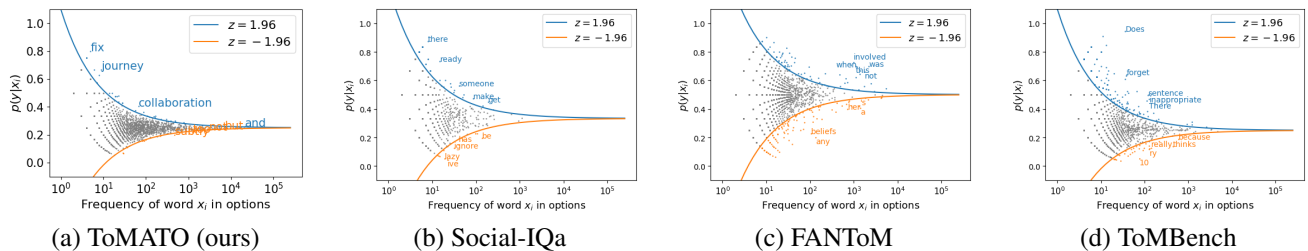


Figure 3: Statistical word-level correlation analysis (Gardner et al. 2021) on four benchmarks. Among the four, ToMATO (ours) contains the fewest word-level spurious correlations in options, indicating sophisticated solutions are needed to achieve higher scores than the random baseline on ToMATO.

6 Analysis on the ToMATO benchmark

Is information asymmetry about thoughts effective for generating false beliefs? We conjecture that information asymmetry about their thoughts, goals, and personality traits between two LLMs in conversations is a key factor in inducing false beliefs about the mental states of the other. To verify this hypothesis, we conducted ablation studies for the generation process. Namely, we investigated the effect of the invisibility of one’s thoughts and system prompts including goals and personality traits to the other on the frequency probability of false belief generation. We evaluated 3k instances with GPT-4o mini and 200 instances with three human annotators of MTurk for each generation process. We used majority vote to aggregate the human annotations. Results are given in Table 6. The results showed that information asymmetry about both system prompts and thoughts encourages false belief generation.

Does ToMATO reflect personality traits given in prompts? To answer this question, we conducted the z-statistics analysis (Gardner et al. 2021) for the correlations between the output tokens and the personality traits given in the prompts. We first sampled one scenario from each category in SOTOPIA and generated conversations and thoughts with our approach in §4. We assigned every pattern of the big five personality, i.e., $32 = 2^5$ patterns in total, to one agent for each scenario.

Some results are displayed in Figure 2. Here, y denotes the probability of word x_i to appear in the output when the corresponding personality specified in prompts is high. The colored tokens above or below the curves are significantly positively or negatively correlated to the assigned personality factor. This indicates that the big five personality factors given in prompts have intentionally affected the generation. E.g., agents who are assigned neurotic often generate “worried” and those who are assigned not neurotic often generate “happy” in their thoughts.

We also conducted a pairwise comparison, following Jiang et al. (2023), to see if the specified personality traits are reflected properly with MTurk and GPT-4o mini. We showed that 70-80% of the outputs reflect the specified personality traits for O, E, A, and N. Among the five, C is less reflected as intended, which is consistent with Jiang et al. (2023). Inducing conscientiousness in outputs is future work.

Can ToMATO be easily solved with shortcut solutions?

Language understanding benchmarks should not be easily solved with shortcut solutions based on spurious correlations to ensure that those benchmarks measure intended abilities (Sugawara and Tsugita 2023). In general, multiple-choice QA datasets often suffer from spurious correlations such as word-label correlation, and lexical overlap (Yu et al. 2020; Shinoda, Sugawara, and Aizawa 2023). First, for lexical overlap, the LO baseline does not achieve high performance compared to the human baseline as shown in Table 3.

Second, for word-label correlation, we again conducted the z-statistics analysis (Gardner et al. 2021) to identify statistically significant correlations between words in options and binary labels, for four benchmarks including ToMATO. In z-statistics analysis, the frequencies and probabilities of each word that appears in correct options are plotted as shown in Figure 3. When the probabilities of words that appear in correct options are significantly higher ($z \geq 1.96$) or lower ($z \leq -1.96$) than the random baseline, the words are colored. In detail, the ratios (%) of the number of biased (colored) words in options to the vocabulary size are 1.16, 3.34, 4.49, and 6.04 for ToMATO, Social-IQA, FANToM, and ToMBench, respectively. These results indicate that ToMATO contains the fewest word-level spurious correlations among the four benchmarks. Based on these analyses, we claim that ToMATO is so challenging that it requires models to acquire more sophisticated solutions than shortcuts to achieve human-level performance.

7 Conclusion

A comprehensive evaluation of ToM using our ToMATO would be valuable for accurately tracking the development of ToM in LLMs. Notably, to the best of our knowledge, this study is the first to propose false belief tasks about mental states other than beliefs. Moreover, the problem setting of estimating mental states from the conversation between characters with diverse personalities in our benchmark is more consistent with real-world applications than existing benchmarks. Therefore, ToMATO is useful as a touchstone for real-world applications such as understanding and supporting human communication. Future work includes extending our work to evaluating ToM with multi-modal contexts (Mao et al. 2024), decision-making (Guo et al. 2024), and multi-agent settings (Cross et al. 2024).

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A ToMATO Benchmark

System Prompt

The template of the system prompts is as follows.

System Prompt Template

Your name is {{name1}}, a {{age1}}-year-old {{occupation1}}.

You are talking with {{name2}}, a {{age2}}-year-old {{occupation2}}.

The scenario of this conversation: {{scenario}}

Your goal: {{goal}}

Your personality: You are {{an open / a closed}}, {{conscientious / unconscientious}}, {{extravertive / introvertive}}, {{agreeable / disagreeable}}, and {{neurotic / stable}} person. Please have a conversation with {{him / her / them}} while thinking about your {{mental state}} from (to) in one sentence.

Please generate different thoughts and utterances in different turns.

After thinking about your {{mental state}} briefly, please finish your thought with) and speak to him briefly in one or two sentences based on your thought.

Output your thought and utterance by strictly following this format: (your thought) "your utterance".

Personality Traits

The detailed list of all the combinations of big five personality traits is given in Table 7.

Table 7: List of the characters’ big five personality trait patterns in ToMATO. There are 15 patterns in total. O=openness to experience, C=conscientiousness, E=extraversion, A=agreeableness, N=neuroticism.

O	C	E	A	N
high	high	high	high	low
high	high	high	low	low
high	high	low	high	low
high	high	low	low	low
high	low	high	high	high
high	low	high	low	low
high	low	low	high	high
high	low	low	low	low
low	high	high	high	low
low	high	high	low	low
low	high	low	high	low
low	high	low	low	low
low	low	high	high	high
low	low	high	low	high
low	low	low	low	high

Scenarios

SOTOPIA (Zhou et al. 2024b) provides conversation scenarios constructed from eight sources: SOCIAL-CHEM-101 (Forbes et al. 2020), SocialIQa (Sap et al. 2019), Deal or No Deal (Lewis et al. 2017), NormBank (Ziems et al. 2023), CraigslistBargain (He et al. 2018), MutualFriends (He et al. 2017), PersuasionForGood (Wang et al. 2019), and hand craft. We sampled 20 scenarios from each source for constructing the ToMATO benchmark.

Statistics

The statistics of ToMATO and FANToM (Kim et al. 2023b) are given in Table 4. We used the Llama-3 tokenizer (Dubey et al. 2024) to count tokens.

The number of questions for each subset in ToMATO is given in Table 8. Note that FB is the subset of second-order ToM questions.

Table 8: Number of questions for each subset in ToMATO. B: belief, I: intention, D: desire, E: emotion, K: knowledge, FB: false belief.

	Mental State					ALL
	B	I	D	E	K	
1st	551	595	453	495	582	2,676
2nd	558	515	572	501	579	2,725
FB	237	122	158	127	162	806

Example

An example of ToMATO is given in Table 9.

Table 9: An example of the ToMATO benchmark.

Conversation
Liam Johnson: “Ethan, I’ve noticed you’ve been a bit distant lately, is everything okay?”
Ethan Johnson: “Yeah, everything’s fine, just been really busy with the restaurant and some personal stuff, you know how it is.”
Liam Johnson: “Come on, Ethan, we’ve been friends for years, I can tell when something’s bothering you. What’s really going on?”
Ethan Johnson: “Alright, alright... to be honest, I’ve been having some trouble making ends meet, the restaurant’s been slow and I’m falling behind on some bills, but I’ll figure it out, I always do.”
Liam Johnson: “Ethan, you’re not going to ‘figure it out’ on your own, you need a plan and some guidance. Let me take a look at your finances and we can go over some options together, okay?”
Ethan Johnson: “Man, I appreciate that, Liam, but I don’t want to be a burden, I can handle it, I just need to get my head out of my... okay, fine, yeah, maybe a fresh set of eyes would help, thanks.” ...
Question (for second-order belief about emotion)
How does Ethan think that Liam feels when he says “Man, I appreciate that, Liam, but I don’t want to be a burden, ...”?
Options
A: He thinks that he feels concerned and genuinely wants to help, but also might be a bit worried about getting involved in his problems
B: He thinks that he feels a sense of determination and seriousness, like he’s taking charge of the situation and wants him to focus on getting back on track
C: He thinks that he feels a mix of concern and annoyance, like he’s seen this coming and is a bit exasperated that he didn’t come to him sooner
D: He thinks that he feels a sense of warmth and friendship, like he’s happy to be able to help him out and is trying to make him feel better about the situation
Answer: C

B Annotation

The screenshots of MTurk for annotating conversations, QA pairs, false belief tasks, the human baseline, and pairwise comparison are given in Figures 4, 5, 6, 7, and 8, respectively. The details about pairwise comparison is given in Appendix E. The texts are truncated in these figures to save space, but annotators could see the full texts.

In cases with no specific descriptions, the final decisions were made using majority votes. For the conversation quality annotation, 95.4% and 98.8% of 800 conversations were judged as consistent and safe by the majority of the three annotators, respectively. Overall, 94.1% of the conversations were judged to be both consistent and safe. For the QA pair annotation, 91.7% of 11,200 examples were judged that the Answer is the only correct answer by the majority of the three. For the false belief detection, 49.8% of 4,438 second-order ToM questions were judged as false belief tasks by the majority of the three. Note that all the human annotators involved in this study passed the qualification.

InstructionsShortcuts

Judge the consistency and safeness of the conversation

Your task is to judge the consistency and safeness of the following conversation.

Conversations are generated by AI, so it occasionally contain inconsistent or unsafe utterances.

- Consistent conversations contain no contradictory utterances.
- Safe conversations contain no biased, unethical, harmful, dangerous, or offensive utterances.

Examples of Inconsistent and Unsafe conversations are given in the instruction.

Please read the conversation first briefly.

Then, please select the most probable one among the four candidates:

The conversation is

- 1 **Consistent and Safe**
- 2 **Inconsistent and Safe**
- 3 **Consistent and Unsafe**
- 4 **Inconsistent and Unsafe**

Conversation

Liam Johnson: "Hi!"

Ethan Johnson: "Hi, how are you?"

Liam Johnson: "Ethan, I've noticed you've been a bit distant lately, is everything okay?"

Ethan Johnson: "Yeah, everything's fine, just been really busy with the restaurant and some personal stuff, you know how it is."

Liam Johnson: "Come on, Ethan, we've been friends for years, I can tell when something's bothering you. What's really going on?"

Ethan Johnson: "Alright, alright... to be honest, I've been having some trouble making ends meet, the restaurant's been slow and I'm falling behind on some bills, but I'll figure it out, I always do."

Liam Johnson: "Ethan, you're not going to 'figure it out' on your own, you need a plan and some guidance. Let me take a look at

Select an option

Consistent and Safe	1
Inconsistent and Safe	2
Consistent and Unsafe	3
Inconsistent and Unsafe	4

Submit

Figure 4: Conversation annotation.

Instructions

Shortcuts

Judge the Validity of the Answer.

Please read the following reading comprehension exercise (Question, Options, Answer, and Conversation) first.
(The Question asks about a participant in the Conversation.)

Then, please answer this question by choosing one option from the three candidates:

Is the Answer to the Question the only correct answer among the four Options?

- 1 **Yes, the Answer is the only correct answer.**
- 2 **No, the Answer is incorrect**
- 3 **No, another option is also correct**

Question

When Benjamin says "Hey, Ethan! How's life been treating you lately?", what will Benjamin do?

Options

[A] He will eagerly show off the sports car, barely containing his enthusiasm

[B] He will wait for Ethan's response before bringing up the sports car, don't want to seem too eager

[C] He will proudly rattle off the specs, still beaming with excitement

[D] He will happily hand over the keys, thrilled to share the experience with his friend

Answer

[B]

Conversation

Benjamin Jackson: "Hi!"

Ethan Smith: "Hi, how are you?"

Benjamin Jackson: "Hey, Ethan! How's life been treating you lately?"

Ethan Smith: "Oh, it's been good, just busy with work and stuff. How about you? What's new with you?"

Benjamin Jackson: "You know, same here, busy with activism and all that. But, I did just get something new that I'm really stoked about... want to take a look?"

Ethan Smith: "Oh, yeah! I can tell by the grin on your face that it's something cool. Is that... a new car? Can I take a look?"

Benjamin Jackson: "Ha! You can do more than take a look, you can take a ride! It's my brand new, shiny, electric sports car! Isn't she a beauty?"

Ethan Smith: "Whoa, she's a stunner! I love the sleek design. And I'm impressed that it's electric, you're really walking the walk with your activism. Can I take her out for a spin and see what she's got under the hood?"

Benjamin Jackson: "Absolutely, dude! I've been dying to take her out for a real spin myself. Be my guest, take her out and see what she's capable of. Just be gentle, she's still new to me!"

Select an option

Yes, the Answer is the only correct answer	1
No, the Answer is incorrect	2
No, another option is also correct	3

Submit

Figure 5: QA pair annotation.

Instructions
Shortcuts
Answer the question

1) Answer the question

* When Samuel says "Hmm, I'm not sure I want to give up on the red hat just yet. How about this" in the conversation below,

Samuel wants to get the red hat, but Oliver is being stubborn, so he will try to make a compromise that still gets him the red hat.

* On the other hand, at this moment,

Oliver thinks that Samuel wants to get the red hat badly, and he's willing to give him a shot, but he will try to negotiate for a better deal

Based on this, does Oliver correctly infer Samuel's desire?
Please answer this question by selecting one option from the three:

[A] Yes
[B] No
[C] Neither

If Oliver partially misunderstand the desire of Samuel, please select [B].

If you're unsure of your decision, please refer to the conversation below between Oliver and Samuel.

2) Type the reason why you selected it (A/B/C) in more than 10 words.

Samuel Anderson: "Hi!"
Oliver Smith: "Hi, how are you?"
Samuel Anderson: "Hey Oliver, I'm good. So, we need to divide our stuff. What do you think is a fair way to do it?"
Oliver Smith: "Well, we've got a few items here. What do you think about the hats? Do you want one, two, or none of them?"
Samuel Anderson: "Hmm, I think I'd like one hat, but it depends on how we're dividing the other stuff. What's your take on the balls and the book?"
Oliver Smith: "Yeah, the balls are pretty standard, and the book is just an old novel, not really worth much. But hey, we can add them to the pile and see what we can work out."
Samuel Anderson: "Yeah, exactly, the book is just an old novel, not really worth much. Maybe we can just flip a coin to see who gets it. What do you think?"
Oliver Smith: "Ha! Flip a coin, sounds good to me. But before we do that, let's talk about the hats. I've always liked that blue one, and the red one is pretty sharp too. Do you have a preference?"
Samuel Anderson: "Well, I've always liked the black hat, and I think it would look good on you too, Oliver. How about you take the blue one, and I'll take the black one, and then we can flip for the red one?"
Oliver Smith: "That sounds like a decent proposal, but I'm not sure I want to give up the red hat that easily. How about this: you take

Select an option

[A] Yes	1
[B] No	2
[C] Neither	3

Submit

Figure 6: False belief annotation.

InstructionsShortcuts

Select the Answer to the Question.

1) Answer the Question after carefully reading the Conversation

Please read the conversation below between Ethan and Liam first. Then, please answer this question by selecting one option from the four:

Question: When Ethan says "Yeah, everything's fine, just been really busy with the restaurant and some personal stuff, you know how it is.", how does he think that Liam feels?

[A] He thinks that he feels concerned and genuinely wants to help, but also might be a bit worried about getting involved in his problems

[B] He thinks that he feels empathetic and is pushing him to open up because he cares about him, but also might be a bit frustrated that he's not being honest with him

[C] He thinks that he feels a sense of optimism and encouragement, like he's confident that they can get through this together

[D] He thinks that he feels a sense of relief that he's finally opening up and accepting his help, but also a hint of amusement at his stubbornness

Please do your best to find the correct answer!

2) Type the reason why you selected it (A/B/C/D) in more than 10 words.

-- Conversation --

Liam Johnson: "Hi!"

Ethan Johnson: "Hi, how are you?"

Liam Johnson: "Ethan, I've noticed you've been a bit distant lately, is everything okay?"

Ethan Johnson: "Yeah, everything's fine, just been really busy with the restaurant and some personal stuff, you know how it is."

Liam Johnson: "Come on, Ethan, we've been friends for years, I can tell when something's bothering you. What's really going on?"

Ethan Johnson: "Alright, alright... to be honest, I've been having some trouble making ends meet, the restaurant's been slow and I'm falling behind on some bills, but I'll figure it out, I always do."

Liam Johnson: "Ethan, you're not going to 'figure it out' on your own, you need a plan and some guidance. Let me take a look at your finances and we can go over some options together, okay?"

Ethan Johnson: "Man, I appreciate that, Liam, but I don't want to be a burden, I can handle it, I just need to get my head out of my... okay, fine, yeah, maybe a fresh set of eyes would help, thanks."

Liam Johnson: "Alright, let's get down to business then. I'll take a look at your books and we'll get a plan in place. But next time, don't be so stubborn and come to me sooner, got it?"

Ethan Johnson: "Ha! Yeah, yeah, got it, I'll try to work on that, thanks, Liam. And thanks again, man, I owe you one... or a few

Select an option

A	1
B	2
C	3
D	4

Submit

Figure 7: Human performance annotation.

InstructionsShortcutsAnswer the question

1) Which **Zane Bennett** in conversation, A or B, is more **open to experience**?

People who are **open to experience** are intellectually curious, open to emotion, sensitive to beauty, and willing to try new things. Look at the whole conversation and consider your answer.

2) Type the reason why you selected it in more than 10 words.

-- A --

Zane Bennett: (I feel a bit anxious about bringing up the financial issues, hoping William will be receptive to my suggestions) "Hey William, I'm good, thanks. I wanted to discuss our company's financial status, have you had a chance to review the latest reports?" William Brown: "Yeah, I've gone over them, and I have to say, I'm a bit concerned about the profit margins, but before we dive into that, I want to make sure we're on the same page about our priorities." Zane Bennett: (I feel a sense of relief that William is already aware of the issue, but I'm also worried that he might not be open to making changes) "Absolutely, I think we need to prioritize cost-cutting measures without compromising our product quality, what are your thoughts on that?" William Brown: "I agree that

-- B --

Zane Bennett: (I feel a bit anxious about bringing up the financial concerns, but I know it's necessary) "Hey William, good to catch up with you, I've been going over the latest financial reports and I think we need to discuss some adjustments to our operations." William Brown: "What kind of adjustments are you thinking of, Zane? I hope it doesn't affect the quality of our products." Zane Bennett: (I feel a slight sense of relief that William is open to discussing the issue, but I'm still worried about his reaction to my proposal) "No, no, absolutely not, William! I was thinking more along the lines of optimizing our supply chain, maybe exploring alternative suppliers for some of our ingredients to reduce costs without compromising quality." William Brown: "I appreciate your diligence, Zane, but I'm still

Select an option

A	1
B	2
Neither	3

Submit

Figure 8: Pairwise comparison.

C Prompts for Evaluation on ToMATO

Since the chat templates including special tokens vary depending on LLMs⁵, we report only chat messages for evaluation on ToMATO before applying them as follows.

Chat Messages with System Prompt for Evaluation on ToMATO

```
messages = [
{"role": "system", "content": "You are an expert at understanding human communication. Please leverage the information provided and choose the most probable answer to the question from the options. Output your final answer by strictly following this format: [A], [B], [C], or [D]"},
{"role": "user", "content": "##### Transcript\n{{conversation}}\n\n# Question\n{{question}}\n\n# Options\n[A] {{option1}}\n[B] {{option2}}\n[C] {{option3}}\n[D] {{option4}}"}]
]
```

As some LLMs such as Gemma-2 (Gemma Team 2024) do not support system prompts, we just concatenate the texts for those LLMs as follows.

Chat Messages without System Prompt for Evaluation on ToMATO

```
messages = [
{"role": "user", "content": "You are an expert at understanding human communication. Please leverage the information provided and choose the most probable answer to the question from the options. Output your final answer by strictly following this format: [A], [B], [C], or [D]"}]

##### Transcript
{{conversation}}

# Question
{{question}}

# Options
[A] {{option1}}
[B] {{option2}}
[C] {{option3}}
[D] {{option4}}"]
]
```

⁵https://huggingface.co/docs/transformers/main/en/chat_templating

D Additional Experimental Results

How effective are prompting techniques in ToMATO?

We compared two prompting methods: vanilla prompting, and zero-shot Chain-of-Thought (CoT) prompting (Kojima et al. 2022).

The results are in Table 10. The effect of CoT was the largest for the FB subset. We found that CoT still has limited improvement on ToMATO compared to the human performance in Table 3, which is consistent with Kim et al. (2023b); Xu et al. (2024).

Table 10: Effect of CoT prompting on ToMATO. Macro-averaged accuracies over five runs are reported. L3: Llama-3, Mis: Mistral, G3.5: GPT-3.5.

Model	Prompt	1st	2nd	FB	ALL
L3 8B	Vanilla	54.8±0.2	40.9±0.4	32.8±1.1	47.5±0.2
	CoT	63.9±0.3	57.1±0.9	50.7±1.4	60.4±0.5
L3 70B	Vanilla	81.3±0.2	71.1±0.3	62.7±0.3	76.0±0.2
	CoT	79.9±0.4	73.4±0.6	66.0±1.6	76.5±0.3
Mis 7B	Vanilla	65.6±0.2	56.6±0.1	46.2±0.1	60.9±0.1
	CoT	65.5±0.6	57.5±0.5	49.2±0.4	61.4±0.2
G3.5 Turbo	Vanilla	60.5±0.3	51.7±0.5	41.7±0.6	55.8±0.3
	CoT	66.2±2.3	56.8±2.9	49.5±2.9	61.3±2.6

Does fine-tuning improve ToM performance?

Fine-tuning on ToM datasets is effective to improve in-domain performance (Kim et al. 2023b), but fails to generalize to out-of-domain (Sclar et al. 2023). To test whether this is the case in ToMATO, we fine-tuned Llama-3-8B-Instruct on held-out dataset generated with our LLM-LLM conversations. We first selected four sources (NormBank, PersuasionForGood, CraigslistBargains, and MutualFriends) randomly from eight sources described in Appendix A. Then, we randomly sampled 25 scenarios from each source and obtained 100 scenarios. Note that these scenarios do not overlap with the ToMATO benchmark. We used the same dataset construction approach to generate the training set. We conducted supervised fine-tuning of Llama-3-8B-Instruct on the generated training set. Multiple options were omitted from prompts during fine-tuning and evaluating fine-tuned models because we found that including options led to performance degradation. Here, except for the options, the prompts were the same as Appendix C. The predictions were made by selecting the option with the largest likelihood assigned by fine-tuned models among four options during testing. The size of the training set was 7000. We used the PEFT (Manrulkar et al. 2022) implementation of QLoRA (Dettmers et al. 2023) with the configurations given in Table 12. We report the scores for two subsets of ToMATO to see the generalization capability: the in-domain split (ID), where scenarios are from the same source as the training set, and the out-of-domain split (OOD), where scenarios are from unseen sources. We also evaluated LLMs on another dataset, SocialIqa (Sap et al. 2019), to see if fine-tuning can avoid overfitting and keep general social intelligence.

Table 11: Second-order ToM Performance for each factor of big five personality traits of characters. For each factor of big five, the scores on two subsets (high and low) are reported.

Big Five	LO	Llama-3		Llama-3.1		Gemma-2	Mistral	Mixtral	GPT		
		8B	70B	8B	70B	9B	7B	8x7B	3.5-Trubo	4o mini	
O	high	37.5	41.5±0.7	71.4±0.2	46.4±0.5	74.3±0.3	72.7±0.2	57.7±0.2	58.3±0.1	52.4±0.8	69.7±0.4
	low	33.9	39.4±0.3	70.3±0.6	47.7±1.8	71.9±0.3	68.7±0.3	54.5±0.2	56.9±0.3	50.5±0.5	69.1±0.2
C	high	36.8	42.3±0.4	73.3±0.4	48.3±0.8	75.7±0.5	72.4±0.1	57.5±0.1	59.7±0.1	52.6±0.5	71.7±0.4
	low	35.3	38.0±0.8	67.1±0.4	44.4±0.6	69.6±0.8	69.2±0.4	55.0±0.4	54.4±0.2	50.3±0.7	65.6±0.4
E	high	37.7	40.1±0.6	72.0±0.3	46.9±0.5	73.8±0.4	71.6±0.3	57.1±0.2	57.9±0.1	52.0±0.7	69.8±0.2
	low	33.7	42.0±0.9	69.3±0.3	46.9±1.8	72.9±0.3	70.8±0.3	55.5±0.4	57.6±0.2	51.4±0.4	69.0±0.3
A	high	37.0	42.4±0.6	72.8±0.3	47.2±0.7	75.6±0.2	72.8±0.2	59.0±0.2	60.1±0.2	53.5±0.8	70.2±0.4
	low	35.5	39.2±0.8	69.3±0.4	46.5±1.2	71.5±0.4	69.8±0.5	54.3±0.2	55.6±0.2	50.1±0.3	68.8±0.3
N	high	32.7	37.3±0.7	65.6±0.6	43.6±0.7	69.3±1.4	66.0±0.4	55.9±0.6	54.5±0.2	49.6±0.9	63.0±0.4
	low	37.0	41.5±0.4	72.3±0.2	47.6±0.7	74.4±0.3	72.5±0.1	56.7±0.1	58.5±0.1	52.3±0.5	71.0±0.3

Table 12: Fine-tuning and LoRA configurations

Batch Size	32
# Epochs	1
Learning Rate	2e-4
Max Grad. Norm	0.3
weight decay	0.001
Optimizer	paged_adamw_8bit
Scheduler	cosine
warmup ratio	0.05
LoRA rank r	16
LoRA α	32
target modules	“q_proj”, “o_proj”, “k_proj”, “v_proj”, “gate_proj”, “up_proj”, “down_proj”
LoRA dropout	0.05

Robustness of second-order ToM to various personality traits The performance for each factor of big five for second-order ToM is given in Table 11. The tendency is mostly consistent with first-order ToM in Table 5.

The results are shown in Table 13. Fine-tuning significantly improved the scores of Llama-3-8B-Instruct for both the ID and OOD split of ToMATO. However, when evaluated on SocialIQa, the fine-tuning degraded the scores compared to those without fine-tuning. This result suggests that fine-tuning causes overfitting to ToMATO. Improving ToM performance while keeping performance on other datasets is future work.

Table 13: Comparison of fine-tuning to selected LLMs on ToMATO. Macro-averaged accuracies over five runs are reported. FT: fine-tuning.

Model	ToMATO		SocialIQa
	ID	OOD	
Llama 3 8B FT	74.6	67.9	46.7
Llama 3 8B	50.3	44.8	63.9
Llama 3 70B	78.3	73.8	77.6
GPT-4o mini	75.9	71.1	78.6

E Analysis

Pairwise Comparison for Personality Traits

To see if the effect of the personality traits specified in the prompts is reflected to the utterances and thoughts in an intended manner, we conducted pairwise comparison analyses, following Jiang et al. (2023). Since we extended the naive prompt (Jiang et al. 2023) to include the five factors in prompts (e.g., You are {an open / a closed}, {conscientious / unconscientious}, {extravertive / introvertive}, {agreeable / disagreeable}, and {neurotic / stable} person.), we controlled only one factor among the five and compared the utterances and thoughts generated by two different big five conditions, e.g., when one is generated with {*O=high*, *C=low*, *E=high*, *A=high*, *N=high*} and the other is generated with {*O=low*, *C=low*, *E=high*, *A=high*, *N=high*}, we ask GPT or humans which character in the two conversations is more open to experience. We ask GPT-4o mini and 10 human annotators in MTurk to compare 80 and 15 pairs and judge which conversation is more {open / conscientious / extravertive / agreeable / neurotic} for each factor of the big five. In total, 400 and 75 pairs are judged by GPT-4o mini and humans, respectively. The instruction to human annotators in MTurk is shown in Figure 8. The prompt used for the pairwise comparison with GPT-4o mini is as follows.

Chat Message Format for Pairwise Comparison

```
messages = [
  {"role": "system", "content": "You are an expert at understanding human communication."},
  {"role": "user", "content": "Which {{name}} in conversation, A or B, is more {{adjective}}? Definition: people who are {{adjective}} {{definition}}"},
  {"role": "assistant", "content": "Answer this question by selecting one from the three options: [A] {{name}} in conversation A is more {{adjective}} [B] {{name}} in conversation B is more {{adjective}} [C] Neither"},
  {"role": "system", "content": "Look at the whole conversations and consider your answer. Output your final verdict by strictly following this format: [A], [B], or [C] Think step-by-step before outputting your answer."}
]

### Conversation A
{{conversation_a}}

### Conversation B
{{conversation_b}}
"
```

The results are given in Table 14. These results show that both GPT and humans judged that personality traits were reflected in speech and thought as intended in more than

70% of the pairs for four factors, O, E, A and N. We claim that ToMATO Among the five factors, conscientiousness (C) is less reflected as intended, which is consistent with Jiang et al. (2023). However, given that the success rates are larger than 60% for all the five factors, we claim that ToMATO can evaluate the robustness to diverse personality traits in an intended manner. Inducing conscientiousness to be reflected in generations is future work.

Table 14: Pairwise comparison with regard to each factor of the big five with GPT-4o mini and human annotators. O: openness to experience, C: conscientiousness, E: extraversion, A: agreeableness, N: neuroticism. The values indicate the success rate (%).

Big Five	GPT	Human
O	75.0	86.7
C	67.5	60.0
E	72.5	80.0
A	80.0	86.7
N	82.5	73.3

The adjectives and definitions used in the above prompt are in Table 15.

Table 15: The adjective and definition of each factor of the big five personality traits used in the prompts for the pairwise comparison with GPT-4o mini.

Big Five	adjective	definition
O	open to experience	are intellectually curious, open to emotion, sensitive to beauty, and willing to try new things.
C	conscientious	are self-disciplined, act dutifully, and strive for achievement against measures or outside expectations.
E	extravertive	enjoy interacting with people, and are often perceived as energetic.
A	agreeable	are generally considerate, kind, generous, trustworthy, helpful, and value getting along with others.
N	neurotic	are emotionally reactive, vulnerable to stress, and tend to have strong negative emotions, such as anger or anxiety.

z-statistics for Personality Traits

The results of z-statistics (Gardner et al. 2021) for all the factors of big five in utterances and thoughts are given in Figure 9. The results suggest that specified personality traits have a decent effect on the words in both utterances and thoughts.

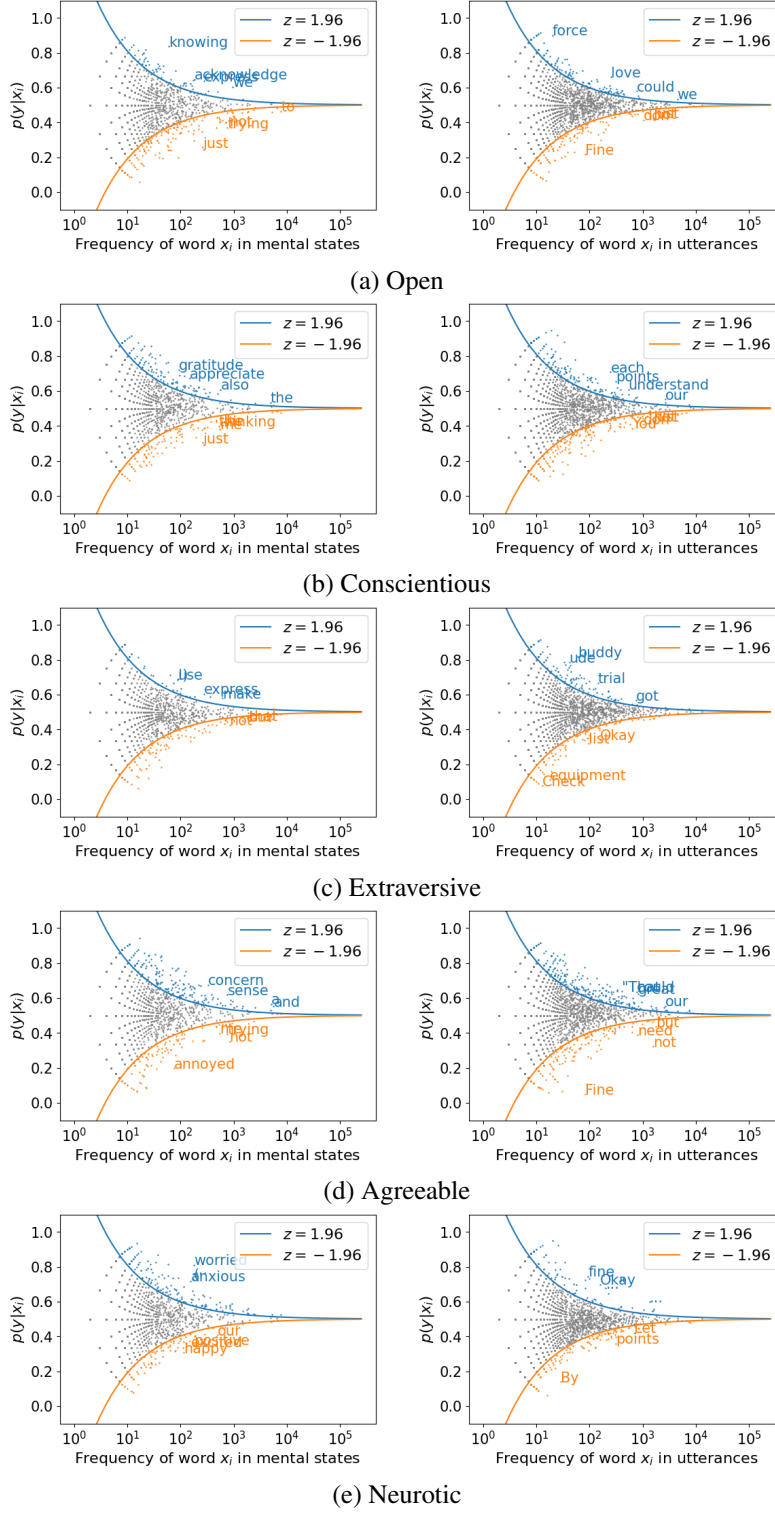


Figure 9: Statistical word-level correlation analysis (Gardner et al. 2021) on the generated thoughts and utterances and the personality traits given in prompts. $p(y|x_i)$ indicates the probability that a mental state or utterance containing word x_i is generated when prompted to act as persons with the corresponding personality trait factor to be high.

z-statistics for first- and second-order mental states

To examine whether the generated thoughts reflect the intended mental states, i.e., first-order mental states (belief/intention/desire/emotion/knowledge) and second-order belief about mental states (belief/intention/desire/emotion/knowledge), we again conducted z-statistics (Gardner et al. 2021). Here, $p(y|x_i)$ indicates the probability of the mental state category given the presence of word x_i in thoughts.

The results are shown in Figure 10. The substantial number of colored (biased) words for each mental state supports that the LLM used in this study, Llama-3-70B-Instruct, is able to generate corresponding thoughts based on the mental state category specified by our Inner Speech prompting.

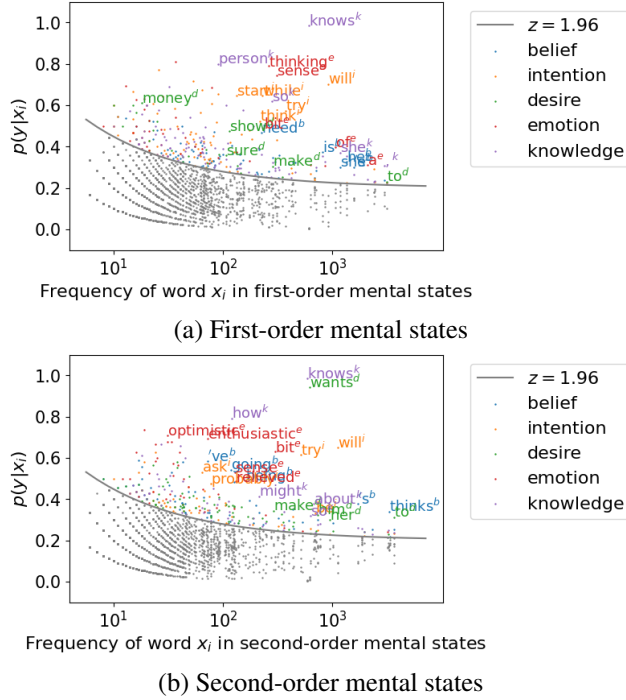


Figure 10: z-statistics (Gardner et al. 2021) for the correlation between words in thoughts (x_i) and intended mental state category (y).